

Smart AgriTech EMS — Documentation

Energy & IoT Management System (EMS) for organizations that monitor field devices, gateways, sensor telemetry, alarms, and analytics.

Documentation index

#	Document	Description
1	System overview & flows	End-to-end journeys, roles, device lifecycle, ingest, alarms
2	Architecture	Components, data stores, deployment topology, security
3	Tech stack	Languages, frameworks, infrastructure choices
4	Application functionality	Feature catalog by role with diagrams
5	Backend	API, services, workers, database, ingest pipeline
6	Web frontend	Routes, pages, state, real-time UI flows
7	Deployment guide	CapRover, VPS options, pricing, CI/CD
8	Database architecture	ER diagrams, all tables, keys, indexes, Redis, data flows
9	CF features implementation	Custom dashboards, access/device groups, facilities, power flow — frontend & backend detail

PDF exports

Formatted PDFs (A4, headers/footers, Mermaid diagrams rendered) live in [pdf/](#) :

PDF	Source
README.pdf	Index
01-system-overview-and-flows.pdf	Flows
02-architecture.pdf	Architecture
03-tech-stack.pdf	Tech stack
04-application-functionality.pdf	Functionality
05-backend.pdf	Backend
06-web-frontend.pdf	Web frontend

PDF	Source
07-deployment-guide.pdf	Deployment
08-database-architecture.pdf	Database
09-cf-features-implementation.pdf	CF features (frontend & backend)
OPTIMIZATION_GUIDE.pdf	Root <code>OPTIMIZATION_GUIDE.md</code>

Regenerate after editing Markdown:

```
cd docs && npm install && node convert-to-pdf.js
```

Repository layout

```
smartagritechapp/
├─ ems/ems-backend/      # Node.js API (Express + Prisma + PostgreSQL)
├─ web_frontend/         # React dashboard (Vite + Tailwind)
├─ app/                  # Flutter mobile app (companion client)
├─ scripts/              # CapRover pack scripts, device simulator, fleet seed
├─ deploy/               # Production env examples
├─ .github/workflows/    # CapRover auto-deploy (backend + frontend)
├─ script.py             # Optional MQTT → HTTP ingest bridge
└─ docs/                 # This documentation set
```

Quick start (local development)

```
# Backend
cd ems/ems-backend && npm install
cp .env.example .env    # set DATABASE_URL, JWT_SECRET, REDIS_URL
npx prisma migrate deploy && npm run seed
npm run dev              # http://localhost:5000

# Web frontend
cd web_frontend && npm install && npm run dev    # http://localhost:5173

# Optional: device simulator
cd ems/ems-backend && npm run simulate:production
```

Default seeded users: `superadmin@ems.com`, `orgadmin@ems.com`, `user@ems.com` (password `Admin@123456`).

Production deployment (summary)

Recommended path: **VPS + CapRover** with four apps (`iotpostgres` , `iotredis` , `iotbackend` , `iotfrontend`). See [Deployment guide](#).

GitHub Actions deploy backend and frontend independently when their folders change on `main` / `master` .